TAYYABA ZUBAID

Frontend Engineer

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SUMMARY

Frontend Engineering intern at FlyRank and Computer Science undergraduate at Qarshi University, building AI-assisted, full-stack web applications with React, Node.js, and Express. Experienced in responsive and accessible UI development, REST API design, and applying object-oriented programming and data structures to build functional, user-oriented software solutions.

EXPERIENCE

FLYRANK July 2026 - Present

Frontend Engineering Intern (AI-Assisted Software Development track)

- Built a full-stack, role-based web application using React, Node.js, Express, and JWT authentication, implementing three distinct user roles (Admin, Teacher, Student) with per-record authorization enforced at the middleware level.
- Designed accessible, responsive interfaces with React, Vite, React Router, and React Context, including ARIA live regions and semantic form markup for real-time dashboards.
- Practiced AI-assisted development workflows, running a controlled comparison of prompt specificity backed by automated tests (Node's built-in test runner) and manual verification, catching a real data-persistence bug in AI-generated code.
- Deployed a production full-stack application using Vercel (frontend) and Render (backend), configuring environment-based API URLs.
- Set up a Node.js/Express REST API and Git-based development workflow from scratch, authoring a project rules file (CLAUDE.md) defining coding, validation, and accessibility conventions for AI-assisted development.

EDUCATION

QARSHI UNIVERSITY 2024 - Present

Bachelor's in Computer Science. Expected Graduation: 2028

PUNJAB GROUP OF COLLEGES 2022 - 2024

Intermediate, FSc Pre-Engineering

PROJECTS

Deenify - An Islamic Website Design

- Developed a mobile-responsive web application using HTML5 and CSS3, focusing on clean layouts and a mobile-first interface.
- Implemented multi-page navigation across five modular pages (Home, About, Features, FAQs, Contact) to improve usability and maintainability.

FinTrack - An Investment Dashboard

- Developed a comprehensive financial dashboard using Java Swing with dynamic graphs, custom table renderers, and a modular navigation system.
- Implemented core application logic using Object-Oriented principles, including an observer design pattern for real-time UI synchronization and a multi-currency profit/loss calculator.

Pokemon Info Card - A Pokemon Encyclopedia

- Developed a dynamic Pokémon encyclopedia using JavaScript and PokeAPI, featuring real-time data fetching for 1,000+ entries.
- Designed a responsive, themed UI/UX with generation-based filtering, interactive stat visualizations, and seamless pagination.

Disaster Evacuation Path Planner - A Console-Based Dashboard

- Architected a real-time disaster evacuation engine in C++ using Dijkstra's Algorithm and dynamic graphs, achieving sub-10-second path recalculation.
- Implemented a Segment Tree to process environmental hazard data with O(log N) efficiency, enabling instantaneous rerouting as conditions evolved.

Weather Dashboard - A Dynamic Weather App

- Developed a dynamic Weather Dashboard using HTML5, CSS3, and JavaScript with real-time weather visualization for major Pakistani cities.
- Implemented adaptive UI/UX elements, including day/night icon switching, along with automated temperature unit conversion and hourly data refreshes.

Tomzora - An Image Gallery

- Developed a dynamic image gallery using vanilla JavaScript and the Unsplash API, with real-time search across 1,000+ photos and advanced filtering.
- Designed a fully responsive, multi-themed UI/UX with 5 color schemes and a mobile-optimized navigation menu adapting from 360px to 1920px+ displays.

Hybrid Sorting Algorithm - A Desktop App

- Built a Python desktop app visualizing five sorting algorithms in real time using CustomTkinter and Tkinter Canvas.
- Designed a generator-based architecture decoupling sorting logic from the animation layer, enabling both live visualization and headless benchmarking via Matplotlib.

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, JavaScript, SQL

Frontend: HTML5, CSS3, React, Vite, React Router, Responsive & Accessible Web Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, bcrypt

Frameworks & Libraries: Java Swing, Bootstrap, React Context API

Data Structures & Algorithms: Arrays, Linked Lists, Trees, Graphs, Dijkstra's Algorithm, Segment Tree

Object-Oriented Programming: OOP principles, Design Patterns (Observer)

Tools & Platforms: Git, GitHub, VS Code, Vercel, Render, Cursor

CERTIFICATES

- **Introduction to Programming with Python and Java** - Specialization Course by Penn Engineering and Coursera
- **Intro to Web Development** - by AKTI
- **Work Smarter with Microsoft Excel** - by Microsoft and Coursera